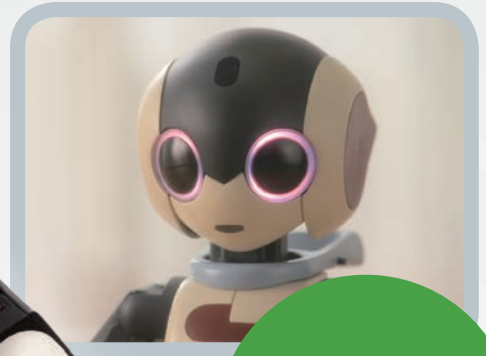
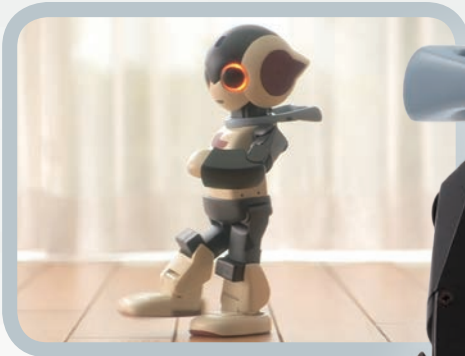


Build your own **Robi**i

Pack 18



Stages 67-70:
Complete your
Robi!



Also with this pack – your
Living with Robi guide!

D'AGOSTINI
MODEL SPACE

Build your own **Robi**

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PLAY. ITEMS MAY VARY FROM THOSE SHOWN. THERE MAY BE
COLOUR VARIATIONS OWING TO THE MATERIALS USED.

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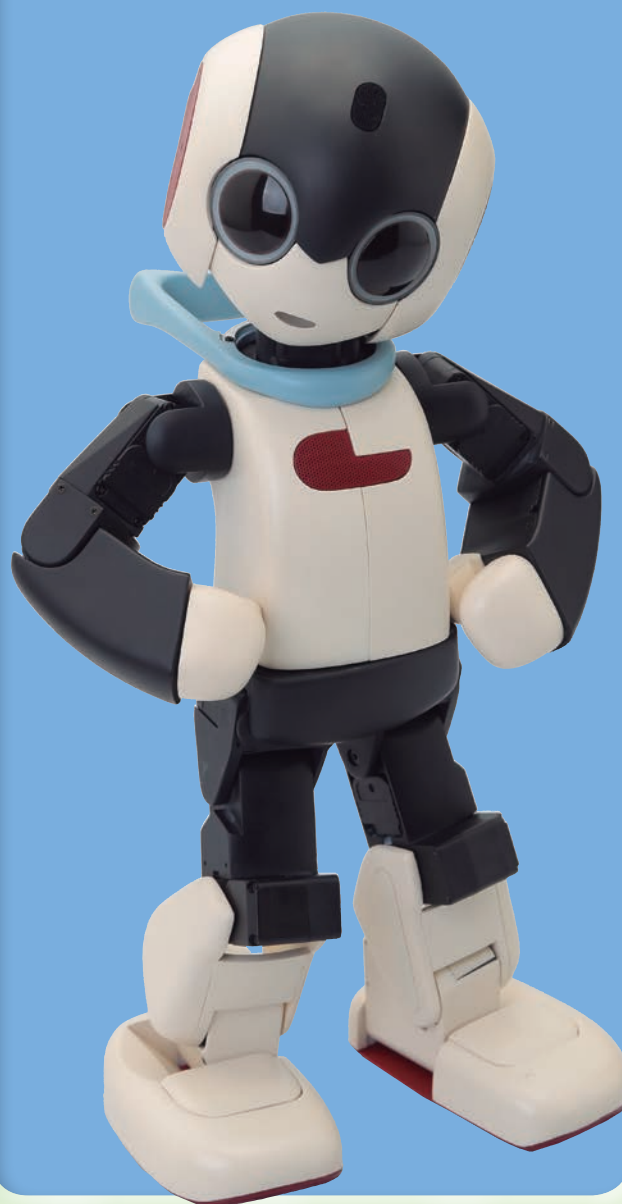
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STAGE 67: ROBI'S VOICE-RECOGNITION BOARD

In this stage, you add another vital electrical component: the voice-recognition (VR) board that is fitted in Robi's head.

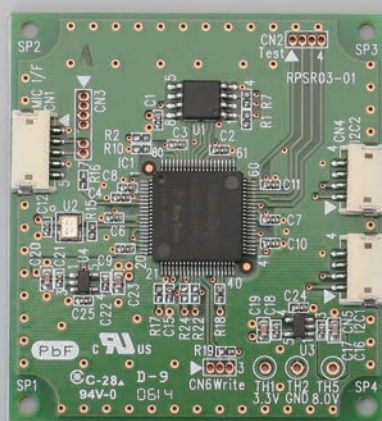
After the CPU, the next most important of Robi's electronic components are the voice-recognition and infrared boards, supplied in this stage and the next stage respectively. These are both built into Robi's head frames, and are stacked on top of one another onto the head base fitted in

the previous pack. The work in this stage is straightforward enough, but again, care must be taken when fitting the VR cables, as they are designed specifically to fit only one way round in their sockets. And, again, always handle exposed electronics very carefully!

PARTS TO BE ASSEMBLED



1



2

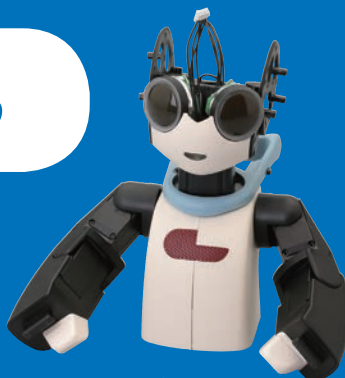


YOUR PARTS

- 1 Voice-recognition (VR) board
- 2 Mounting posts

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Body and head assembly
(Stage 66)

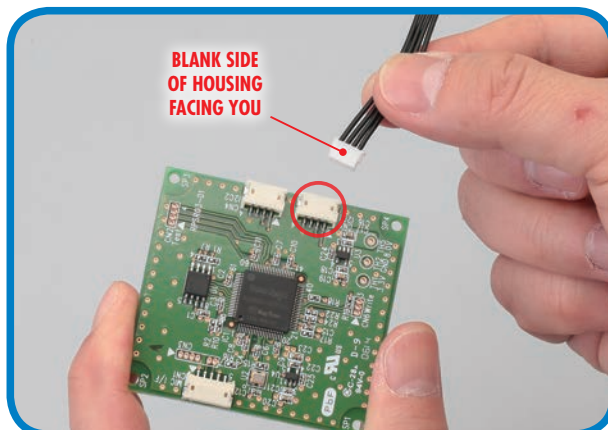


M2 x 5mm pan-head screw
(Stage 64)

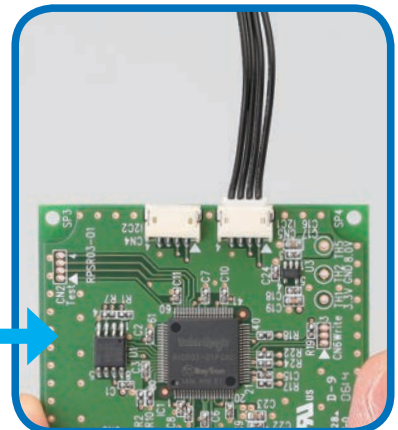
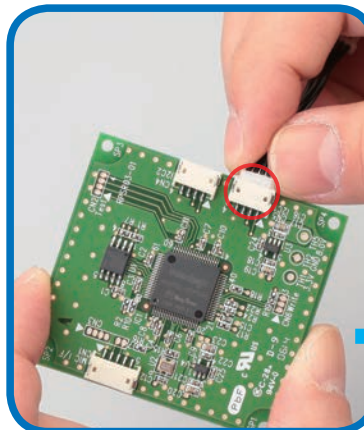


VR cable
(Stage 66)

WIRE THE VR BOARD

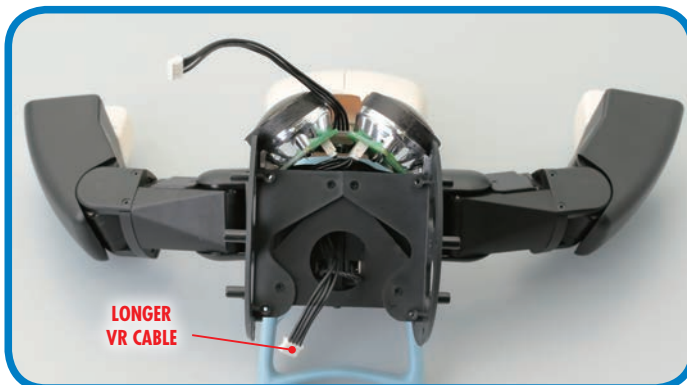


- 1 Line up the 6cm-long VR cable from Stage 66 to the circled socket on the VR board. The blank face of the connector housing should face you. Hold the VR board by its sides.

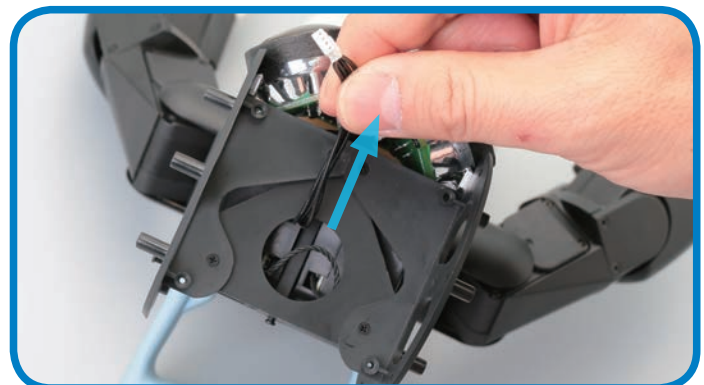


- 2 Making sure the connector is correctly oriented, insert it into the socket. Check that it is set securely in place.

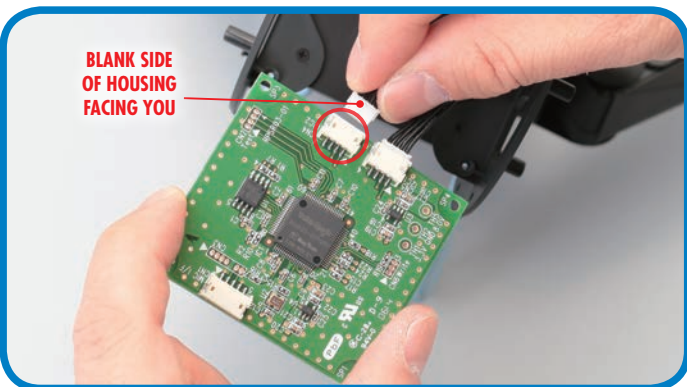
MOUNT THE VR BOARD



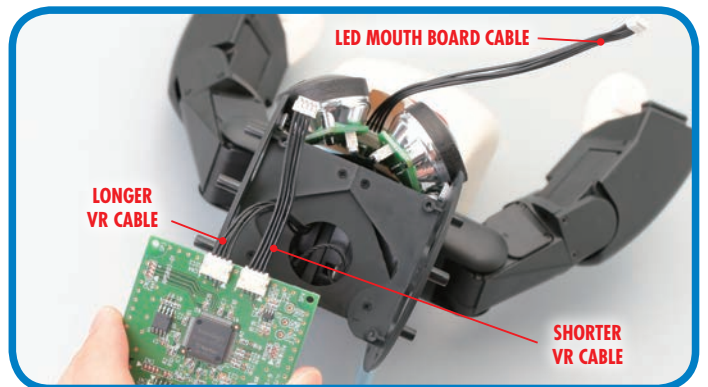
- 3 View Robi's head from above, making a note of the longer VR cable running from the CPU fitted previously.



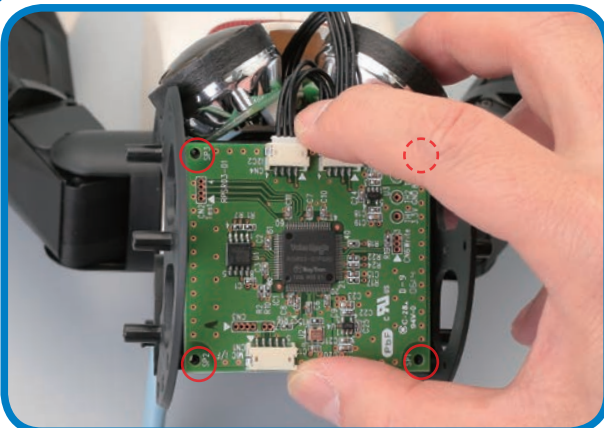
- 4 Gently pull this longer cable towards Robi's front.



- 5 With the longer VR cable's connector facing downwards, as shown, line it up to the free socket on the VR board, adjacent to the one holding the shorter cable fitted in Step 2.



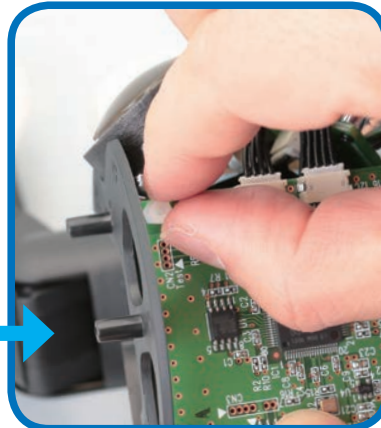
- 6 Fit the cable, double checking each one is where it should be using the photo above.



7 Lower the VR board onto the top of the head base, so its corner holes align with those on the base.



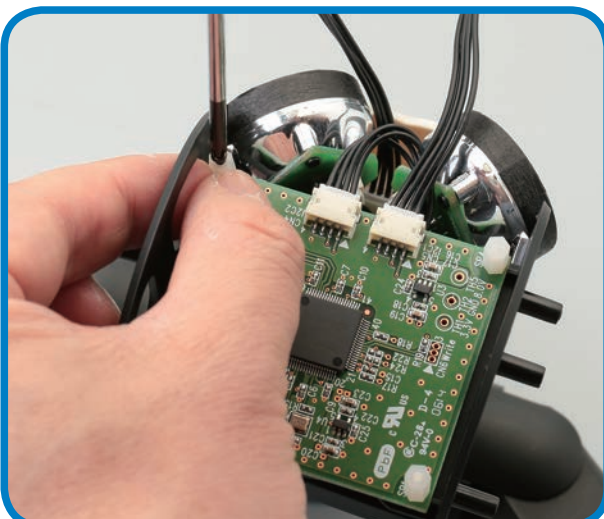
8 Press the first of the mounting posts into one of the holes.



9 Fit all four posts. You should be able to fit these adequately by hand, but as they are hexagonally shaped, a 4mm hobby spanner may be useful.

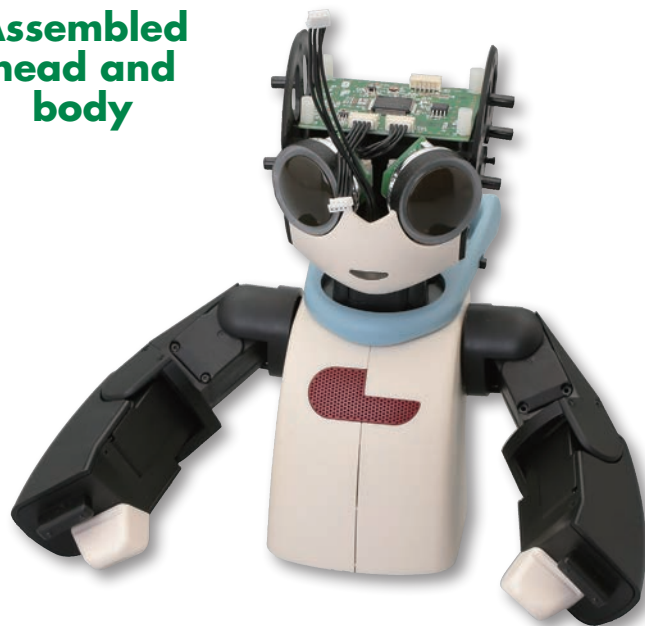


10 Now gently turn the M2 x 5mm screw saved from Stage 64 into the first post. **Be careful not to overtighten it, as the post is made of soft plastic and could be damaged.**



11 Next, holding the post steady, loosen and remove the screw. Repeat Steps 10-11 for all four posts, to cut a thread into them, ready for the coming stages.

**Assembled
head and
body**



STAGE 68: ROBI'S TV REMOTE-CONTROL BOARD

In this stage, you add the TV remote-control board that is wired up to and mounted on top of the VR board set inside Robi's head.

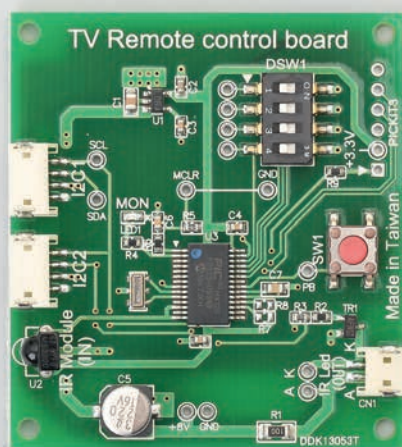
Now that the voice-recognition board has been wired up and fitted to the inside of Robi's head, it is time to add the board that will enable Robi to operate television sets remotely, the TV remote-control (TV/RC) board. The process of installing this is very similar to the work of the previous stage, but

follow the assembly steps closely, as they involve differentiating between very similar-looking cables, which absolutely must be fitted in the right places for Robi to work. It is also important not to overtighten the four screws that secure the TV/RC board, as these enter soft, breakable plastic.

PARTS TO BE ASSEMBLED



1

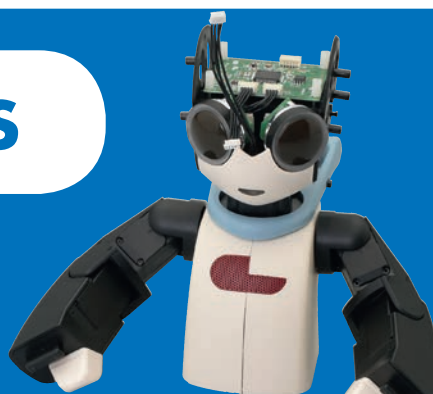


YOUR PARTS

1 TV remote-control (TV/RC) board

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

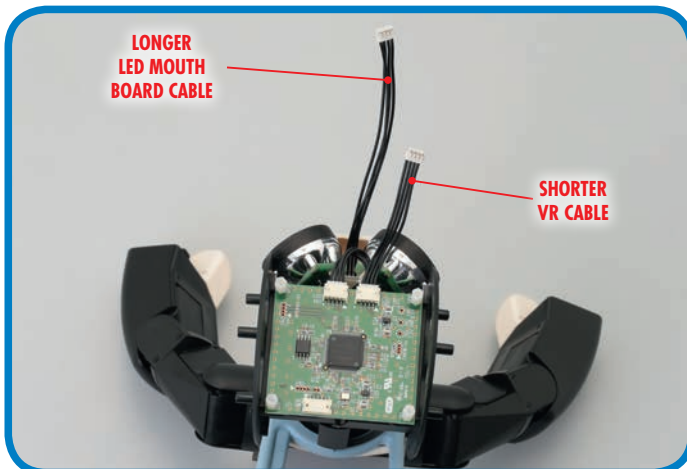


Body and head assembly
(Stage 67)

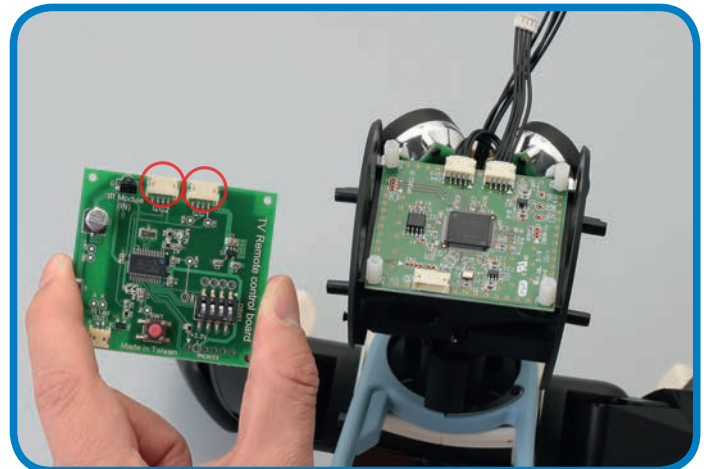


M2 x 5mm pan-head
screws x 4 (Stage 64)

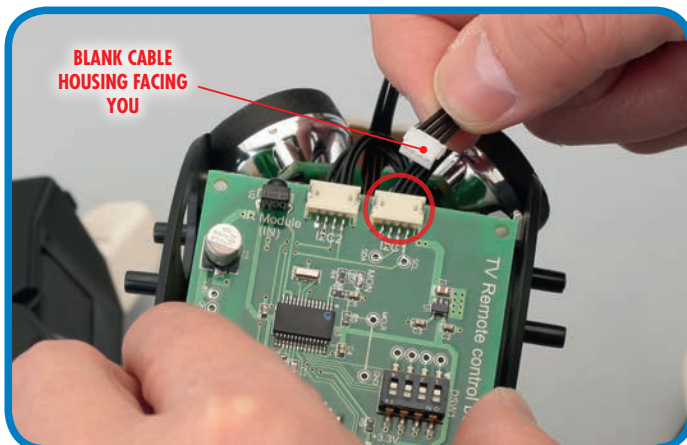
WIRE THE TV/RC BOARD



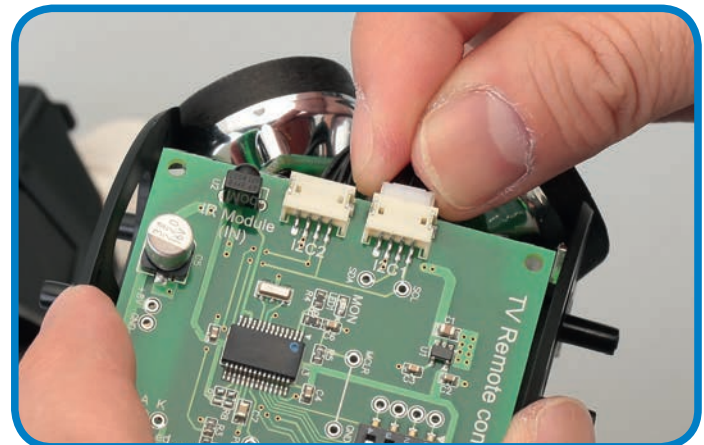
- 1 Look into Robi's head from above and identify the longer LED mouth board cable and the shorter VR cable.



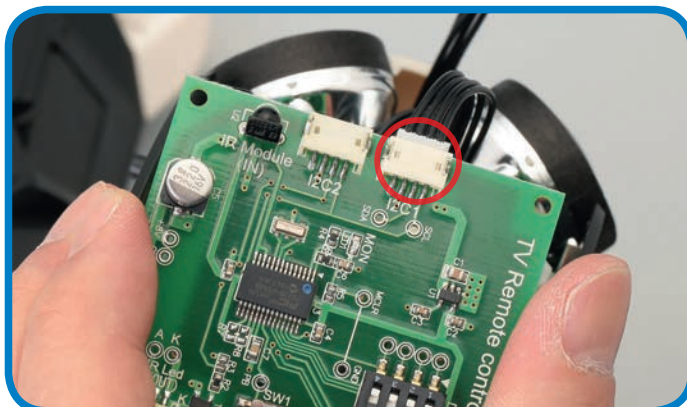
- 2 Line up the TV/RC board, with its two white plastic sockets facing Robi's front, as shown.



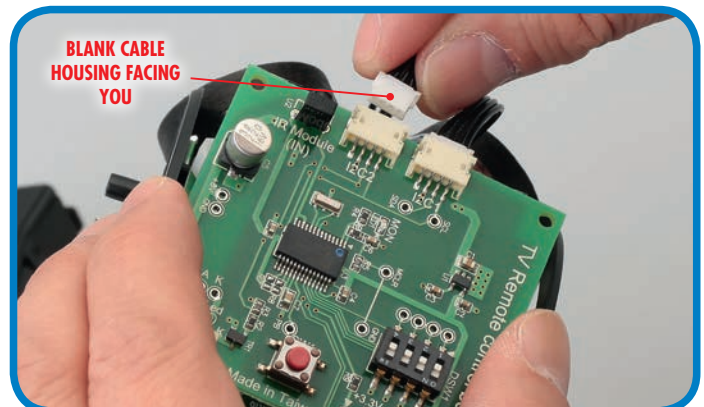
- 3 Take the shorter VR cable and line up its free connector to the circled socket – the one above Robi's right eye. Make sure its blank plastic housing is facing upwards.



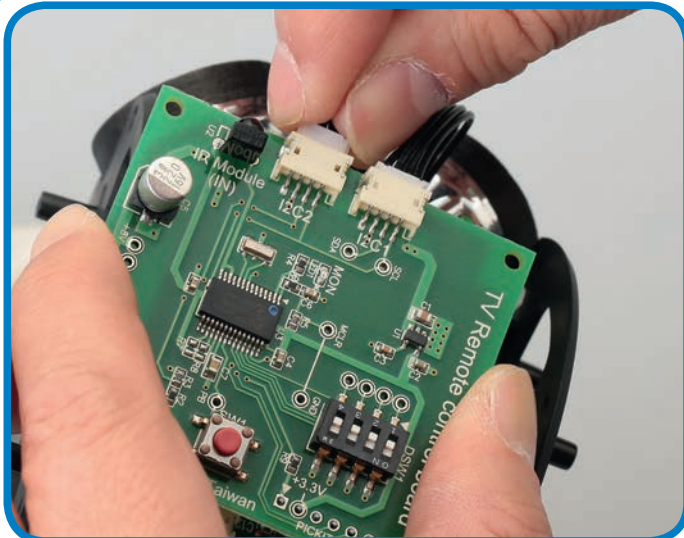
- 4 Fit this into the socket, making sure it is set securely.



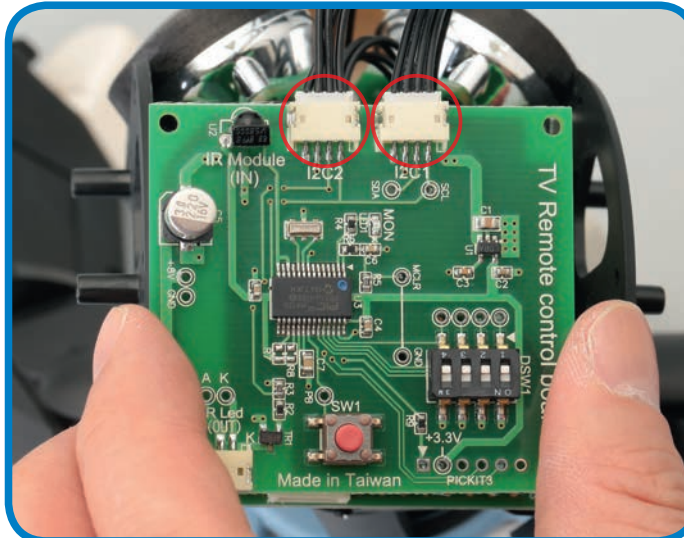
- 5 Check that it is fitted securely before proceeding.



- 6 Now line up the free connector of the LED mouth board cable to the adjacent socket, again making sure its blank plastic housing is facing you.

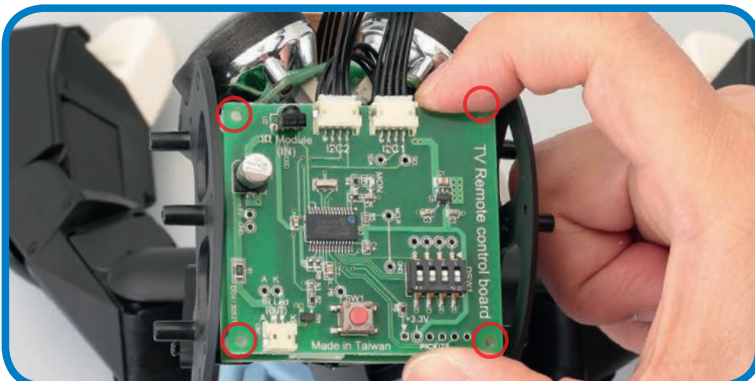


- 7** Set the connector securely into the socket.

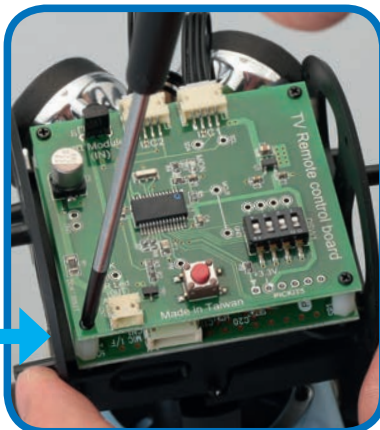


- 8** Check both connections are sound before proceeding.

MOUNT THE TV/RC BOARD

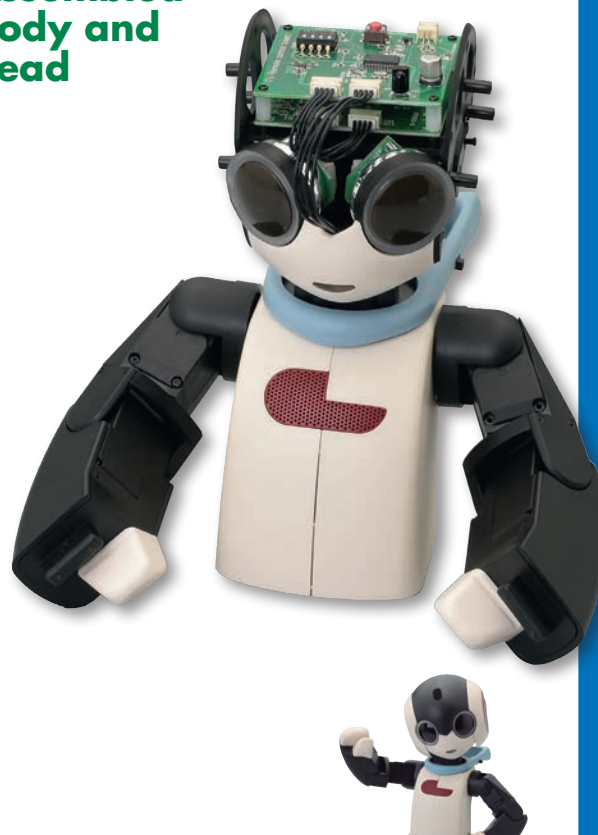


- 9** Lay the TV/RC board on top of the mounting posts fitted to the VR board in the previous stage. The circled holes should align with those cut into the tops of the posts.



- 10** Use four of the M2 x 5mm screws (Stage 64) to secure the board to the posts. Tighten these slowly and gently to avoid damaging the plastic of the posts.

Assembled body and head



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

STAGE 69: JOIN ROBI'S LEGS AND BODY

PARTS TO BE ASSEMBLED

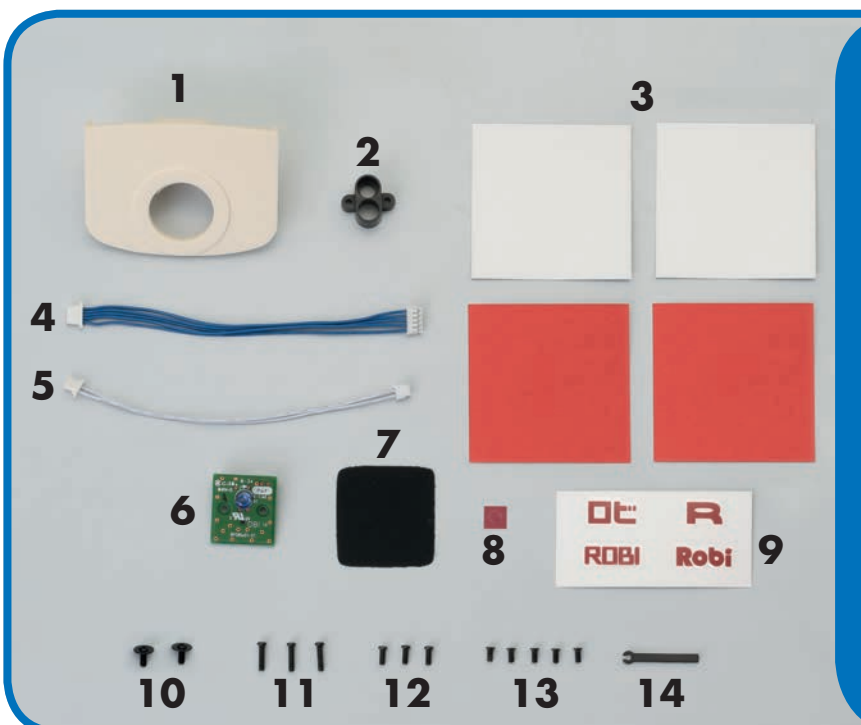


This stage sees you wire and join Robi's legs and body, perform a full-body movement test for the first time, then program Robi's TV remote-control capabilities.

In this, the penultimate stage of the Robi build, you will complete all but the finishing touches on your Robi. This will include connecting up all of

Robi's servos to the CPU, along with the various other cables that join his other electronic components, such as the battery, eye sensors and LEDs.

Then, after a full-body movement test, you will proceed to program Robi to work with your TV (subject to compatibility).



YOUR PARTS

- 1 Body base
 - 2 Infrared LED/mic support
 - 3 Flags (2 x red, 2 x white)
 - 4 Mic cable
 - 5 Infrared LED cable
 - 6 Infrared LED/mic board
 - 7 Foam pad
 - 8 Switch ON sticker
 - 9 Body stickers
 - 10 M2 x 5mm washer screws x 2
 - 11 M2 x 8mm pan-head screws x 3
 - 12 M2 x 5mm pan-head screws x 3
 - 13 M2 x 4.5mm countersunk screws x 5
 - 14 Mini wrench
- (one of each screw-type is a spare)

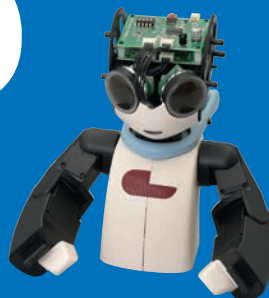
SAVED PARTS



Power distribution board
(Stage 40)



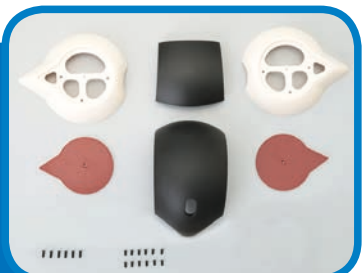
Battery
(Stage 38)



Body and head assembly
(Stage 68)

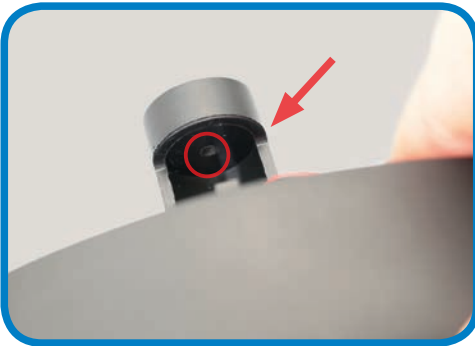


Legs and hips
(Stage 52)



Head components plus screws
(removed Stage 62)

FIT THE BODY BASE



- 1** Hold the legs by the hip base, facing you, and locate the hole in the raised cylinder of the hip base.



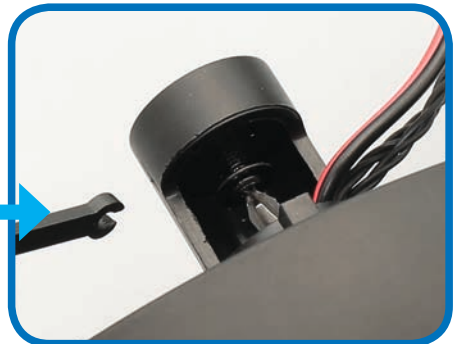
- 2** Use the mini wrench supplied with this stage to hold an M2 x 5mm washer screw, as shown. Fit this screw, shaft side first, into the top of the cylinder from its hollow inside.



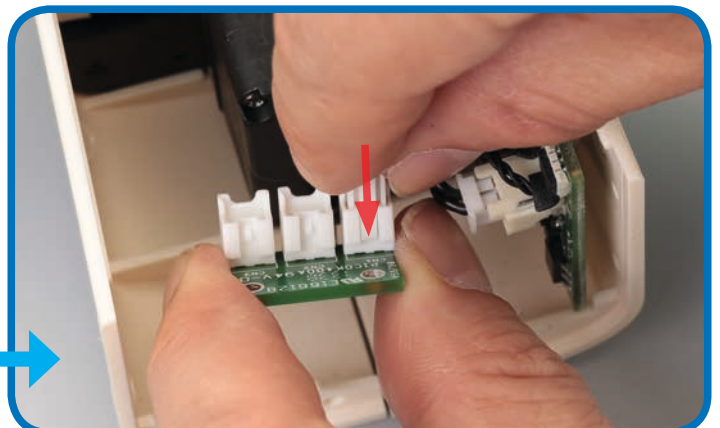
- 3** Now lay the legs out flat and, holding the washer screw in position with the mini wrench, fit the screwdriver through the underside of the hip base to meet the screw head.



- 4** Tighten the screw, most of the way, then when it is set securely enough but not fitted entirely, remove the mini wrench.

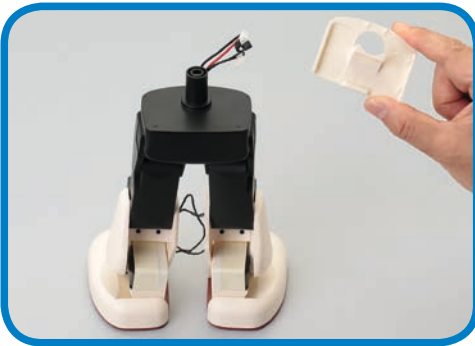


CONNECT THE SWITCH CABLE



- 5** Now, on the body, take the free connector of the power switch and fit it into the socket CN1 on the power distribution board.

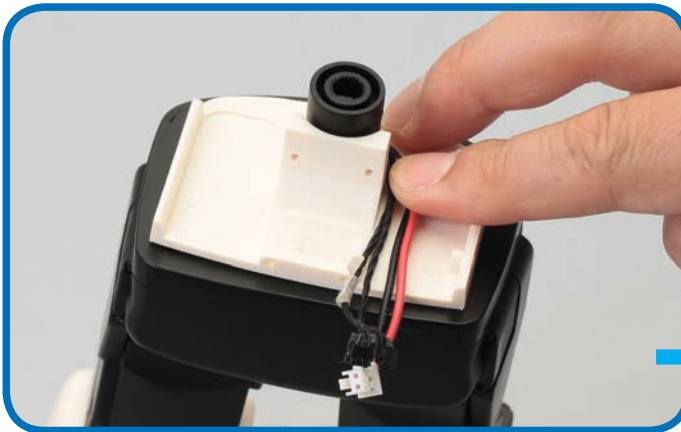
JOIN THE BODY AND LEGS



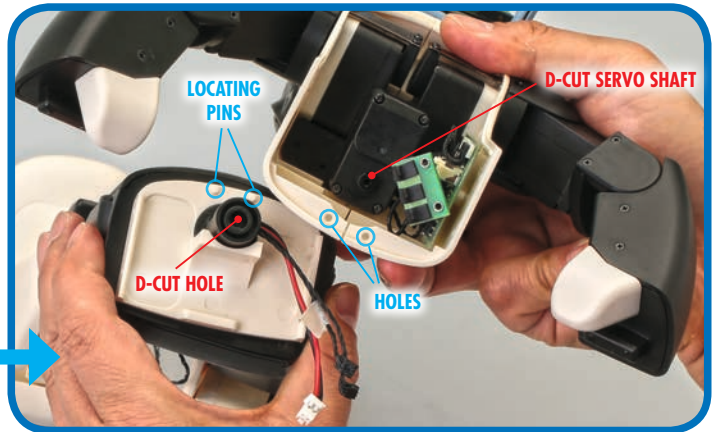
6 Position the legs as shown and line up the body base.



7 Fit the body base to the hip base, feeding the raised cylinder and its cables through as you do.



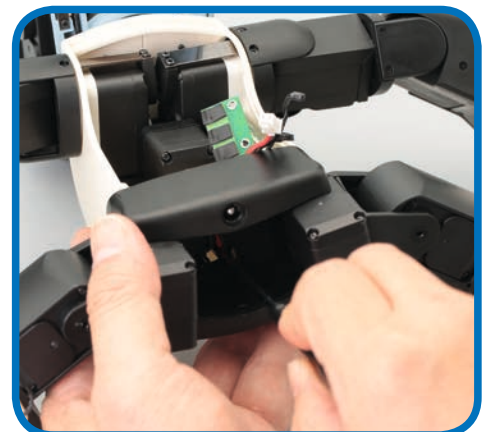
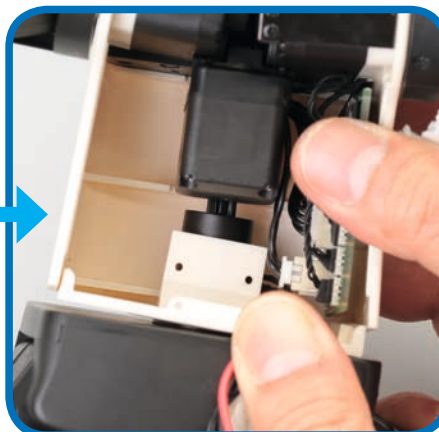
8 With the body base in position, fold the cables around and to the right/back of Robi's legs. Press them down with your fingers to help them remain in position.



9 It is now time to join Robi's body and legs! Before you do, familiarise yourself with how they will join: the back servo's shaft will fit onto the D-cut hole in the top of the hip base's raised post; the two holes at the front of the body will accommodate the locating pins on the body base.



10 Following the guide in the previous step photo, carefully join the parts. **Make sure the D-cut joint between the back servo's shaft and the hip base is set correctly.**



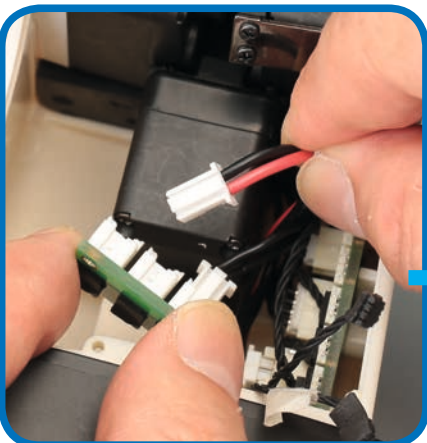
11 Reinsert the screwdriver as in Step 4, and finish tightening the screw.



12 Use two M2 x 4.5mm countersunk screws supplied with this stage to fasten the left side of the body to the body base.

13 Repeat for the right side.

WIRE THE BODY AND LEGS



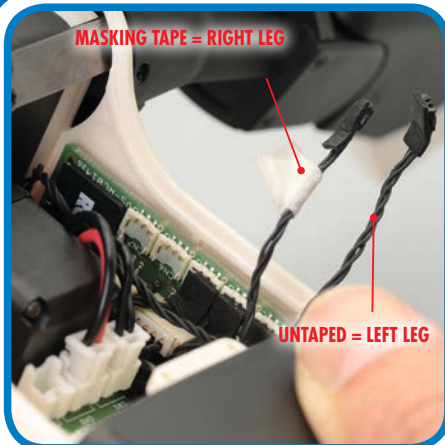
14 Now take the red and black power cord running from the hip base to the body and plug its connector into the central socket on the power distribution board, which is CN2.



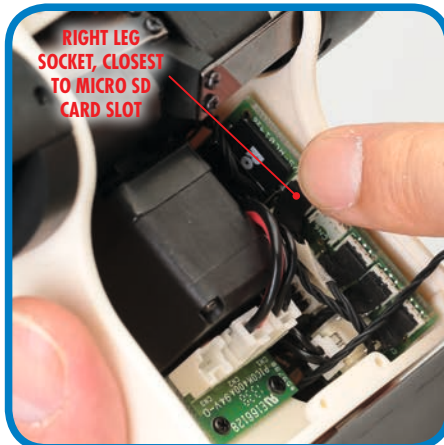
15 Now line up the board to the raised tab moulded to the body base. The holes of both should match up.



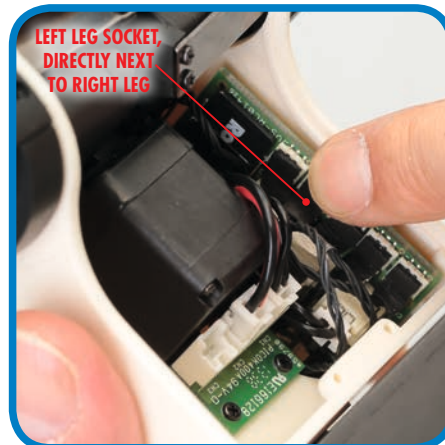
16 Secure the board using two M2 x 5mm screws also supplied with this stage.



17 Now find the servo cable previously marked with masking tape, indicating that it leads to the right leg.



18 Fit the connector of the taped right leg servo cable into its socket on the CPU, which is the one closest to the micro SD card slot.

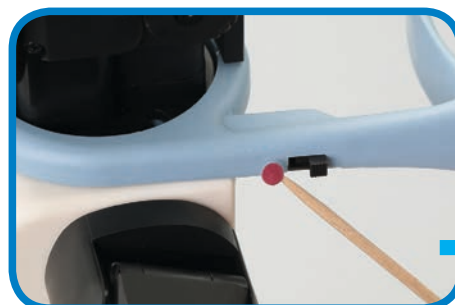


19 Fit the untaped left leg servo cable's connector to the socket directly adjacent to this.

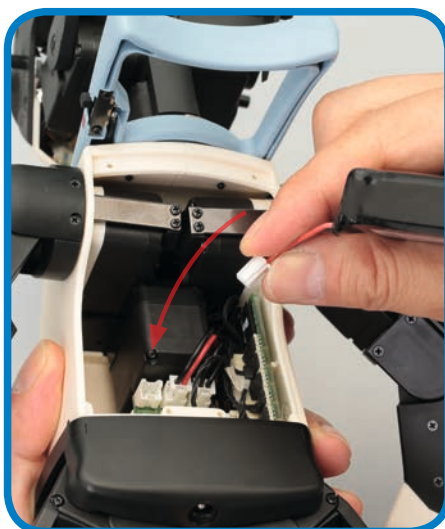
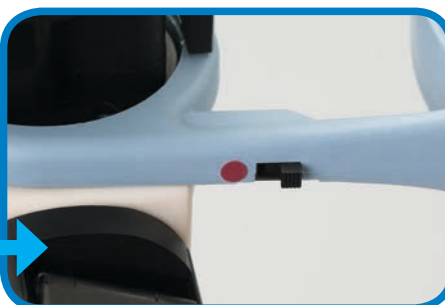
THE SWITCH AND BATTERY



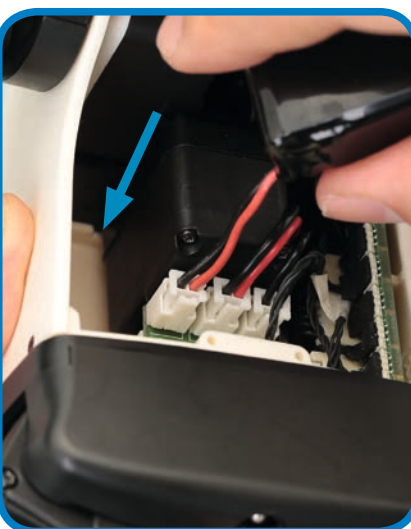
20 Make sure Robi's power switch is in the OFF position.



21 Take the switch's red ON sticker and remove its backing. Then, using a screwdriver tip or toothpick if necessary, apply this to the left of the switch (next to the ON position).



22 Fetch the battery and plug its connector into the last remaining free socket, CN3, on the power distribution board. Then, slide the battery into the space on the left-hand side of Robi's body panels, with its cables running from the bottom of the casing.



23 Press the battery firmly into place.

RECHARGING ROBI



Now you have joined Robi's legs and body, follow the instructions below on how to recharge the battery.

Using the charger

You will charge Robi via the mains adapter in the same way as you have previously. Make sure the power switch is turned OFF before you begin.



1 With the power switch in the OFF position, plug in Robi's mains adapter to his rear hip panel, as you have previously.



2 Turn the mains power ON. The adapter's LED will be red while charging, and turn green when the charge is complete. Switch off and unplug the adapter at this stage.

Using the charging seat

You can also charge Robi using the specially designed charging seat. Again, make sure the power is turned OFF before you do this.



1 Remove the cap from the charging seat's output.



2 Lower Robi onto the seat in the sitting position.



3 Make sure the output enters the charging port properly.



4 Plug the adapter into the port in the rear of the seat.



5 Turn the mains power ON. The adapter's LED will be red while charging, and turn green when the charge is complete. Switch off and unplug the adapter at this stage.

THE FULL-BODY MOVEMENT TEST

ATTENTION!

Hold Robi off the ground by his scarf as you run the test.



- 24** Lift Robi off your work surface by his scarf, then turn the power switch ON.

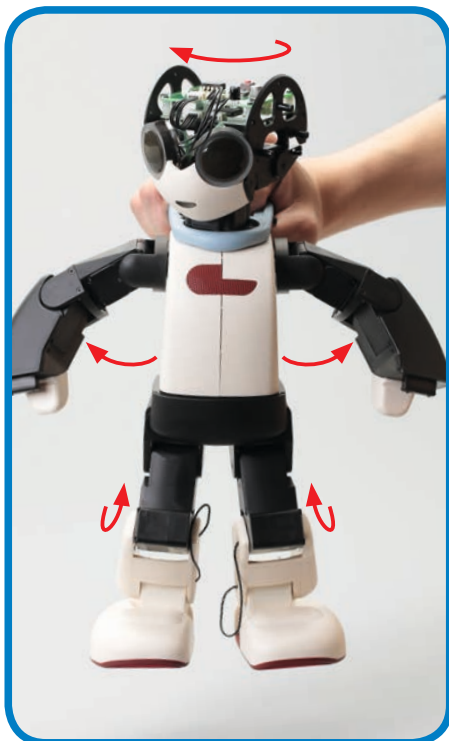
NEUTRAL POSITION



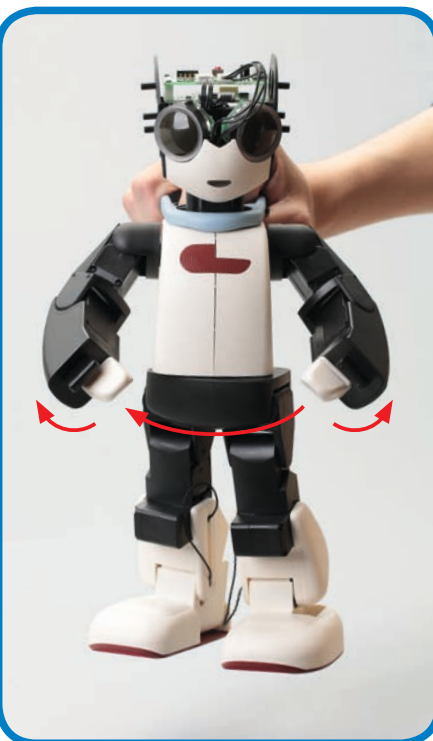
- 25** Robi will assume his neutral position. Observe this closely from the front and sides, as shown above.



- 26** Robi will first extend his arms out at the elbows, spread his legs and tilt his head upwards, then return to the neutral position.



- 27** Next, Robi's legs will push forward, his upper arms lift upwards and his head turn to the right, then return to the neutral position.



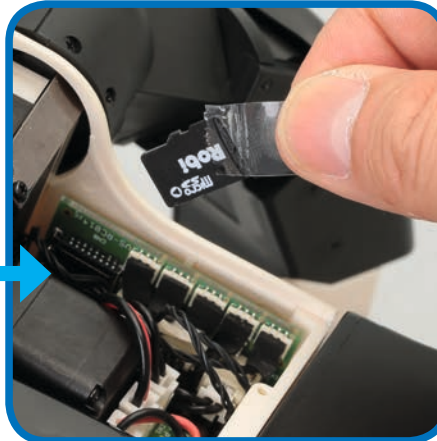
- 28** After that, Robi's arms will extend forward at the shoulder and his hip base rotate to the right, then return to the neutral position.

NEUTRAL POSITION

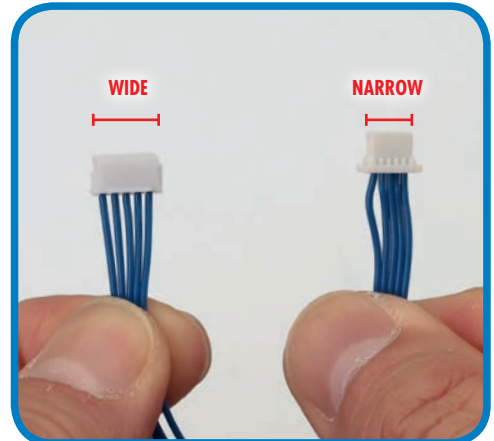


- 29** Now, for the first time, you will see Robi's eyes light up purple, his mouth glow red; his speaker will also say 'OK!' If the test goes as described, switch the power OFF. If anything is amiss, consult the troubleshooting guide of this stage.

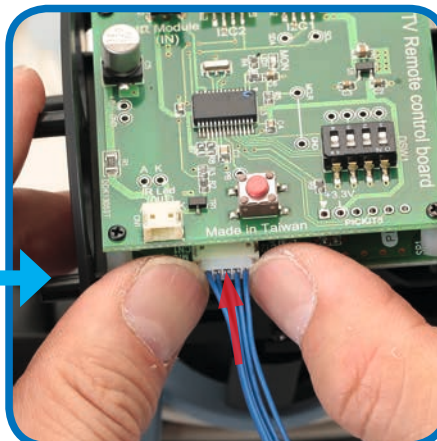
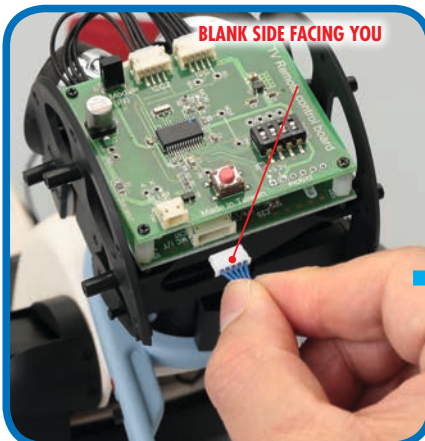
PROGRAM THE TV CARD



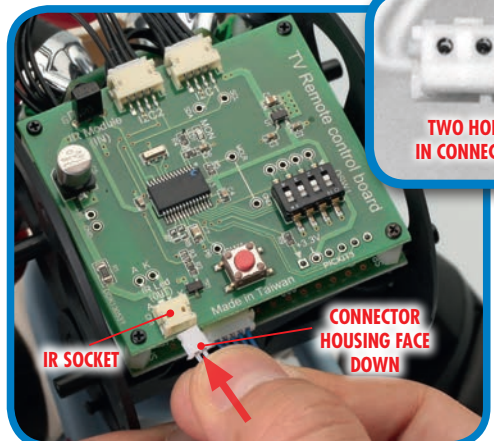
- 30** Remove the test micro SD card from its slot on the CPU. If you find it difficult to do this by hand, use a small length of adhesive tape, as shown.



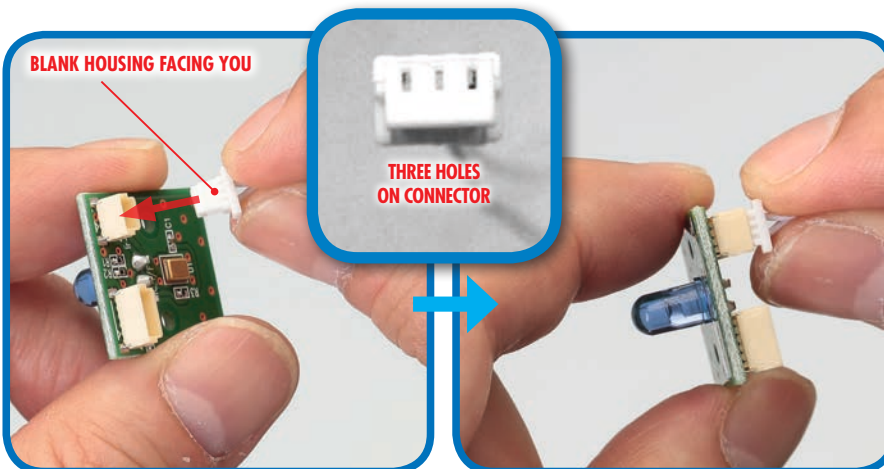
- 31** Take the blue mic cable and identify its wider and narrower connectors.



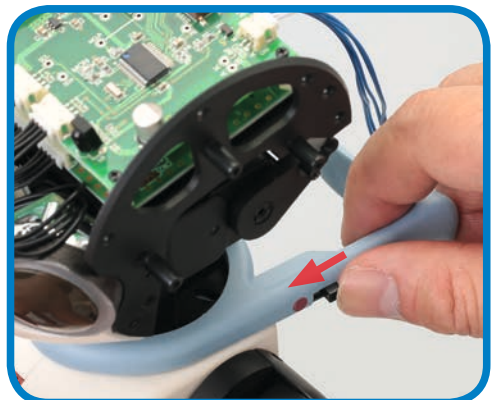
- 32** Insert the wider connector into the lower-set VR board's free socket, at the back of Robi's head. The blank connector housing should face you.



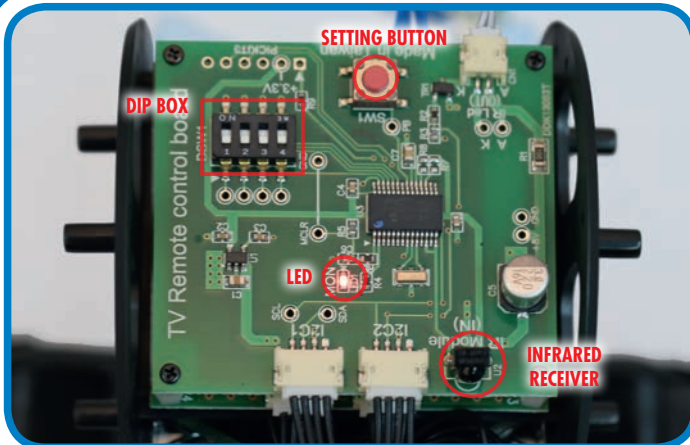
- 33** Now take the infrared/LED cable and identify its smaller, two-pin connector. Fit this into the indicated socket on the TV/RC board. Again, insert the connector face down.



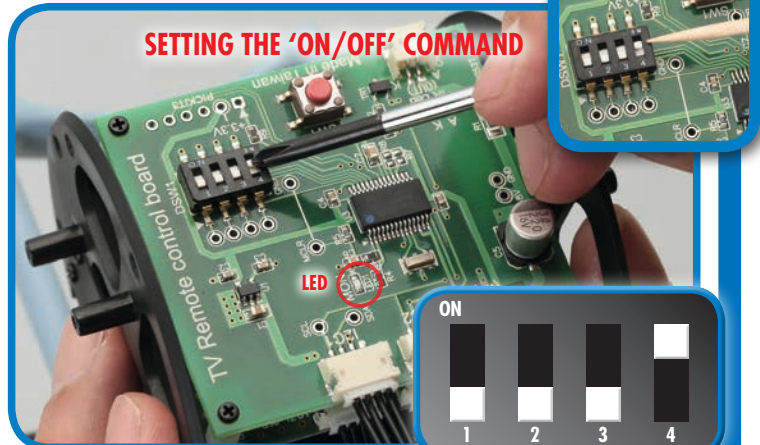
- 34** Slot the larger, three-hole connector at the other end of the cable into socket CN2 on the LED/mic board.



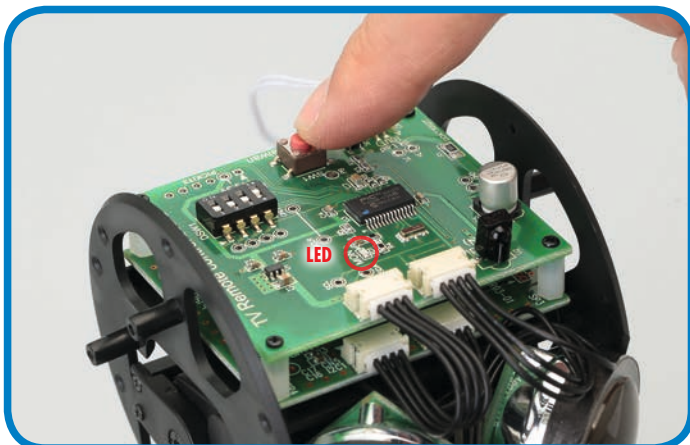
- 35** Turn Robi's switch ON.



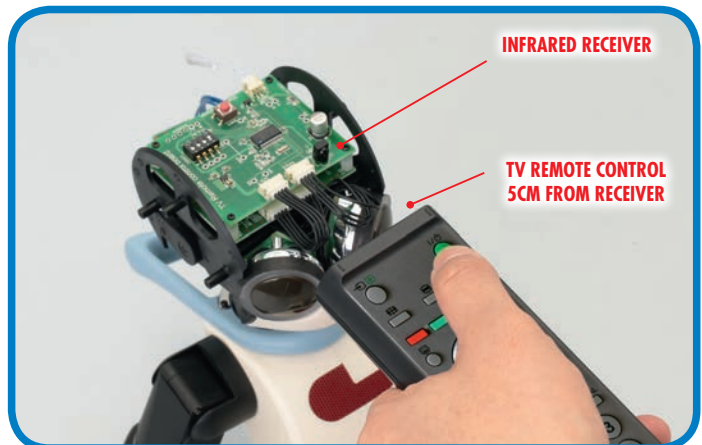
36 Sit Robi down facing you, and familiarise yourself with the components highlighted above.



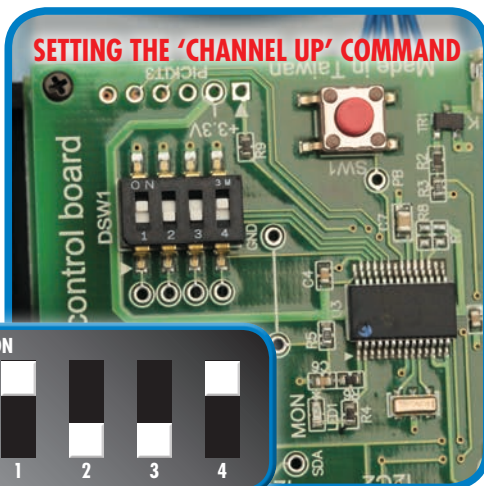
37 You will now begin programming Robi to work with your TV, so begin by using a screwdriver tip or toothpick to push the no. 4 switch on the DIP box to the ON position (the side of the box facing Robi's scarf).



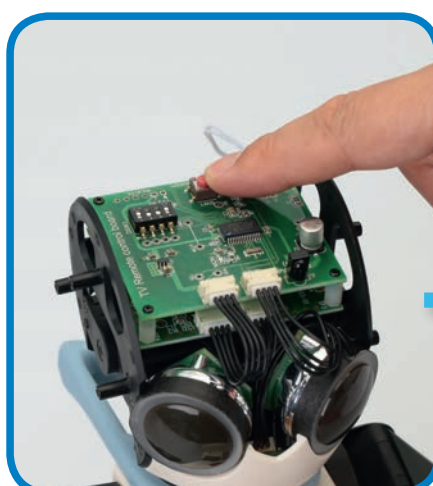
38 Now press the red setting button. The board's red LED should light up.



39 Point your TV remote control at Robi's infrared receiver from about 5cm away, and press the ON/OFF button once. After about a second, the LED on Robi's TV/RC board will go out. The ON/OFF command is set.

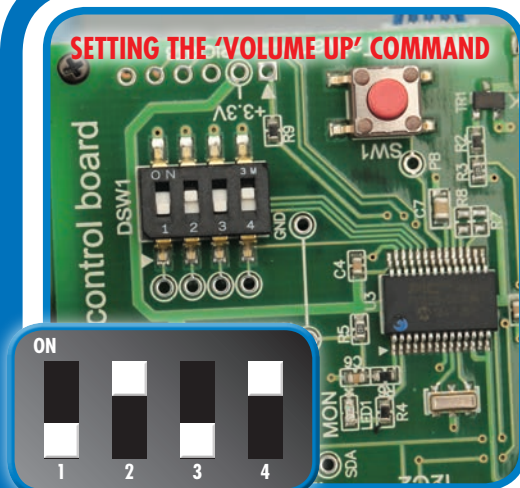


40 Next you will set the 'Channel UP' command. To do this, push the DIP switches no. 1 and 4 to the ON position.

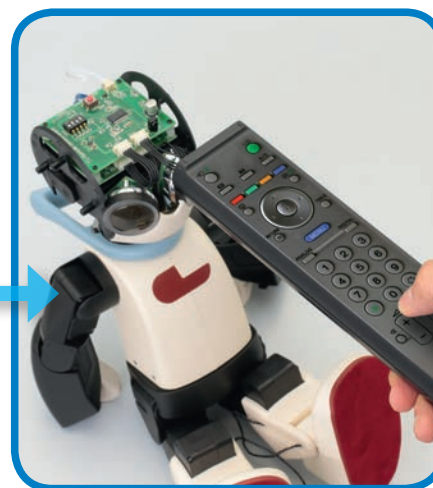
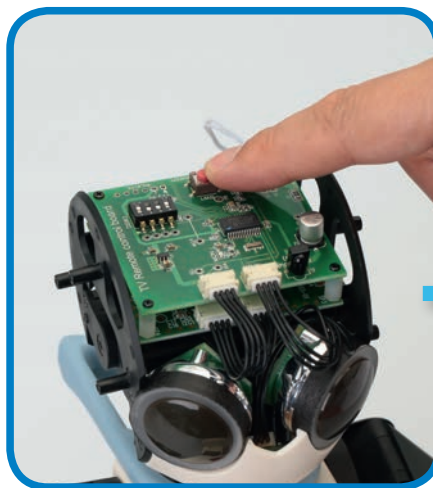


41 Press the setting button again so that the TV/RC board's red LED lights up. Point your TV remote control at Robi's infrared receiver from about 5cm away, and press the 'Channel UP' button once. After about a second, the LED on Robi's TV/RC board will go out. The command is set.

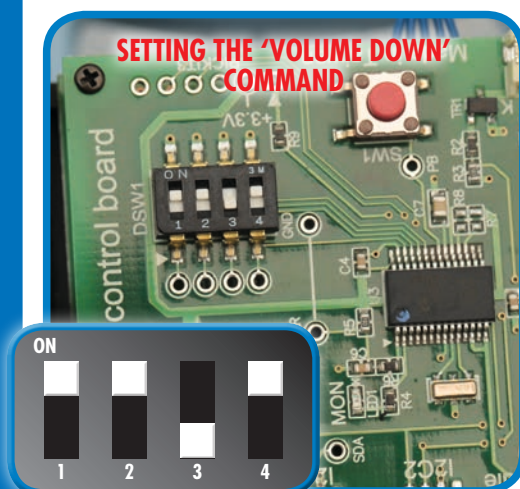




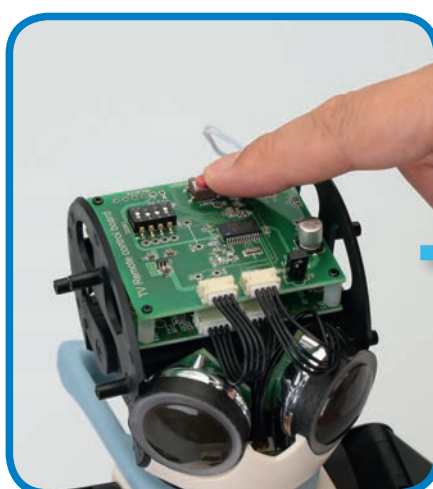
- 42** To set the 'Volume UP' command, push the DIP switches no. 2 and 4 to the ON position.



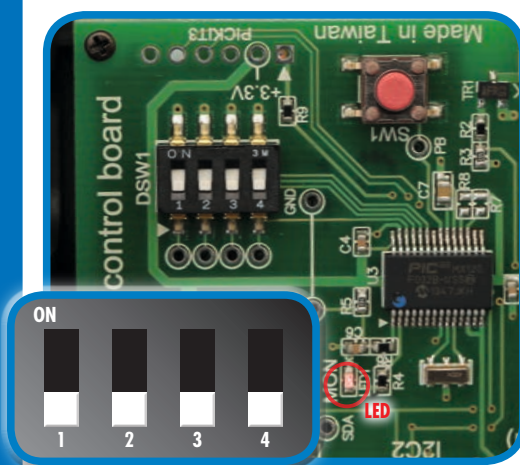
- 43** Press the setting button again so that the IR board's red LED lights up. Point your TV remote control at Robi's infrared receiver from about 5cm away, and press the 'Volume UP' button once. After about a second, the LED on Robi's IR board will go out. The command is set.



- 44** Lastly, you will set the 'Volume DOWN' command. Push the DIP switches no. 1, 2 and 4 to the ON position.



- 45** Press the setting button again so that the TV/RC board's red LED lights up. Point your TV remote control at Robi's infrared receiver from about 5cm away, and press the 'Volume DOWN' button once. After about a second, the LED on Robi's TV/RC board will go out. The command is set.



- 46** Once you are done, return all DIP switches to the OFF position.



- 47** To test your Robi's ON/OFF command, point the LED on the LED/mic board at your compatible TV. Press the setting button once – the TV should turn on. Press the button again to check the TV turns off as well.

TESTING THE 'CHANNEL UP' COMMAND



- 48** Set the combination of DIP switches to the one shown above, then point the LED board at your TV and press the setting button. The channel should change up.

TESTING THE 'VOLUME UP' COMMAND



- 49** Next, set the DIP switches to the combination shown above, then point the LED board at your TV and press the setting button. The volume should increase.

TESTING THE 'VOLUME DOWN' COMMAND

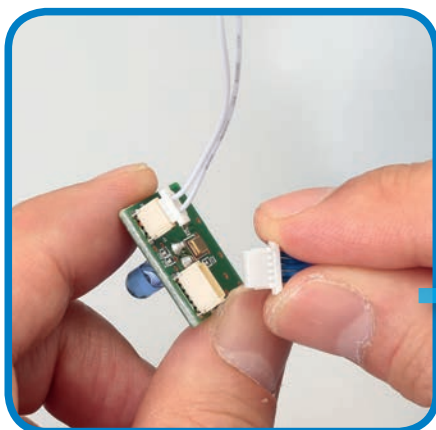


- 50** To test the 'Volume DOWN' command, set the DIP switches to the combination above, then point the LED board at your TV and press the setting button. The volume should decrease.

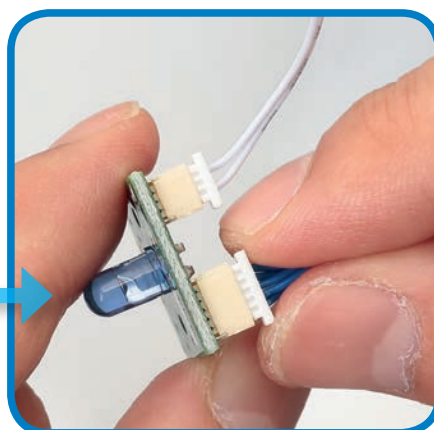


- 51** If all the tests went as described, return the DIP switches to the OFF position. If you need to reset a command, simply repeat that command's steps.

INSTALLING THE LED/MIC BOARD



- 52** Plug the connector of the mic cable into the free socket (CN1) on the LED/mic board. The connector's blank housing should face you.



- 53** Next, take the head cover and fit the foam pad over its central hole from the inside. Make a note of the locations of the two circled screw mounting lugs.



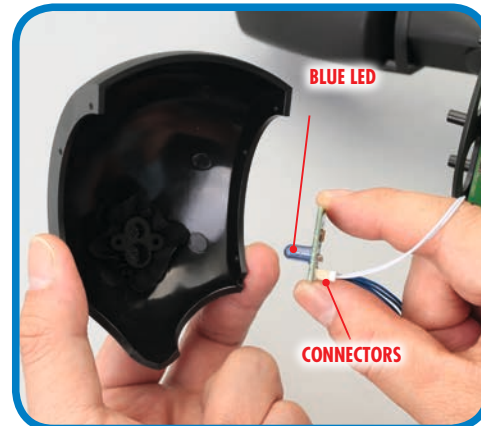
- 54** Position the foam pad like this.



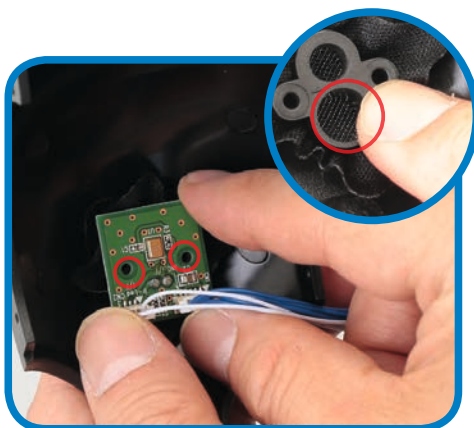
- 55** Take the LED/mic bracket and orient it as shown above left. Its smaller of the two main holes will go at the top, with the two smallest holes on either side to take the screws. Lower the part into position on top of the foam pad.



56 Press the bracket into place. Its smallest side holes should sit on top of the mounting lugs beneath, but with the foam pad still in place.



57 Line up the head top to the LED/mic board, positioned as shown.

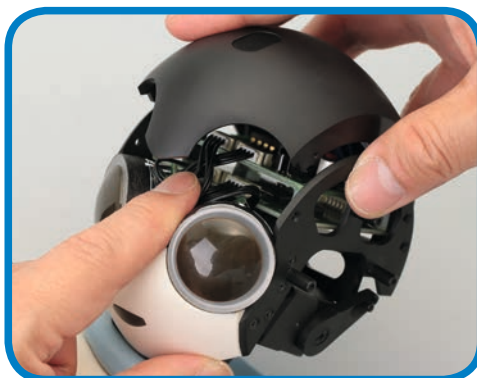


58 Fit the board, with the blue LED sitting in the larger hole at the bottom of the bracket (inset). The circled holes (main image) should align with the smaller ones on the bracket, and the lugs beneath.



59 Secure the board with two M2 x 8mm pan-head screws.

SEALING ROBI'S HEAD



60 Holding the bunch of cables fitted behind Robi's face so that they don't get trapped, lower the head top onto Robi's head assembly.



61 Fix the head top to the side head frames using six of the M2 x 4.5mm countersunk screws removed when you disassembled the head in Stage 62.



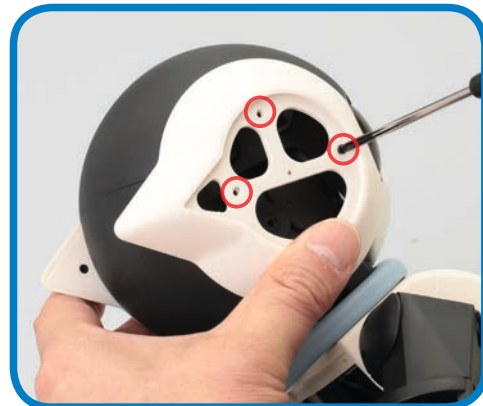
62 Now add the back of head panel.



63 Use four more M2 x 4.5mm countersunk screws to secure this via the circled holes.



64 Next, line up the left ear panel and secure it with three of the M2 x 6mm countersunk screws removed in Stage 62.



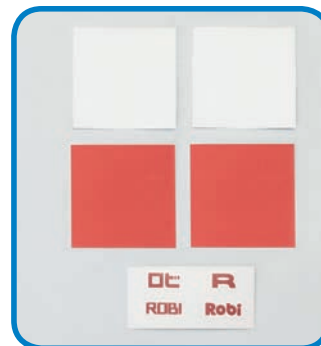
65 Repeat for the right ear panel.



66 Add the ear covers and secure with an M2 x 4.5mm countersunk screw in each.



Store all the unused components carefully!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Assembled head, body and legs



TROUBLESHOOTING

Here, we look at what to do if the movement test carried out in this stage does not go as described.

IF NONE OF THE SERVOS MOVE

If none of the servos move when the test is started, begin by checking the following points:

1. The battery is fully charged
2. The micro SD card is properly inserted
3. The scarf's switch is ON

If all of the above are OK but the problem remains, disconnect all the servos from the CPU and then test each one individually. Doing this will help you establish which servo group (left arm, right arm, neck/head) is faulty and needs attention.

If you find you need to reset any of your Robi's servo ID numbers, make absolutely sure to follow the correct stage's instructions!



IF ONLY SOME OF THE SERVOS MOVE

If some but not all of the servos move, refer to the troubleshooting guides of the relevant stages (eg. Stage 66 for the arms and neck, Stage 52 for the legs, and so on).

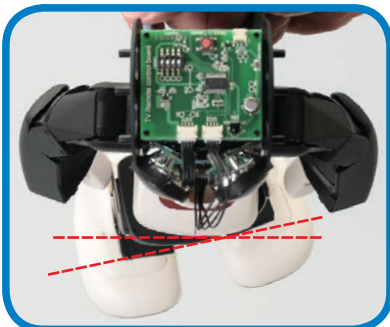
IF THE SPEAKER DOES NOT SAY 'OK!'

If Robi's speaker does not say 'OK!' at the end of the movement test, check the connection made in Step 15 of Stage 57.

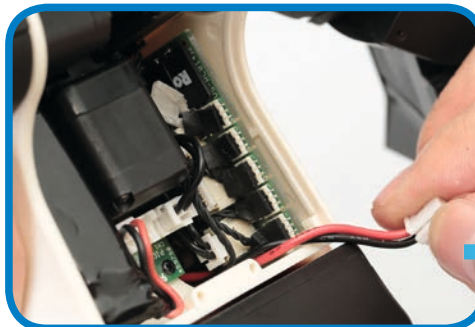
IF THE EYE AND MOUTH LEDS DO NOT LIGHT UP

If the LEDs fitted around Robi's eyes or in his mouth do not light up as they should, check all the related steps of Stages 64-68, making sure all the connections are sound.

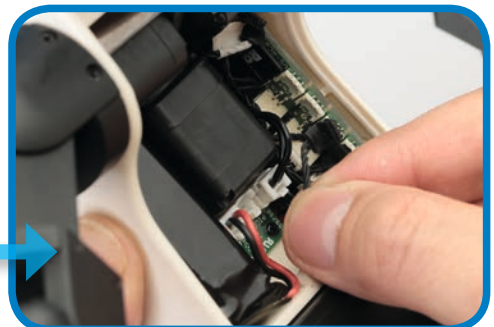
IF THE LEGS ARE NOT POSITIONED PROPERLY



- 1** Check from above to assess if Robi's legs are positioned incorrectly (as seen above).



- 2** If so, proceed to disconnect the power cord and both legs' servo cable connectors. **Use a short tab of masking tape or similar to mark which leg is which.**



- 3** Next, insert the screwdriver into the hole in the underside of Robi's hip base, and loosen the screw there. You do not need to fully remove the screw.



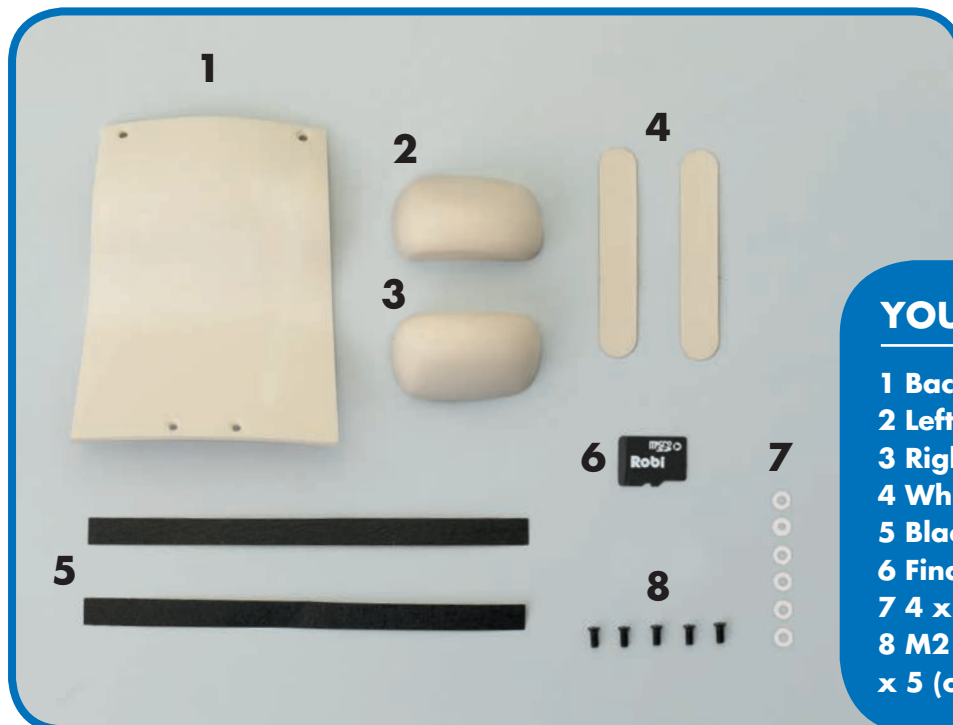
- 4** You can now lift away the body assembly and reset the joint between the back servo's shaft and the cylindrical mount on the hip base. Once this is done correctly, tighten the screw loosened in the previous stage fully. (If the screw came out, see the 'Fit the body' section earlier in this stage.)

STAGE 70: COMPLETE YOUR ROBI!

The final stage of the Robi assembly sees you add his hand and leg coverings, then fit the final micro SD card.

To complete your Robi, all you need to do is add the finishing touches to your assembly. This includes fitting his hand and leg coverings, before inserting the final micro SD card and

sealing in Robi's back. Then, the very final step is to add a Robi sticker onto his chest (you have a choice of designs from the previous stage), and you are ready to bring Robi to life!

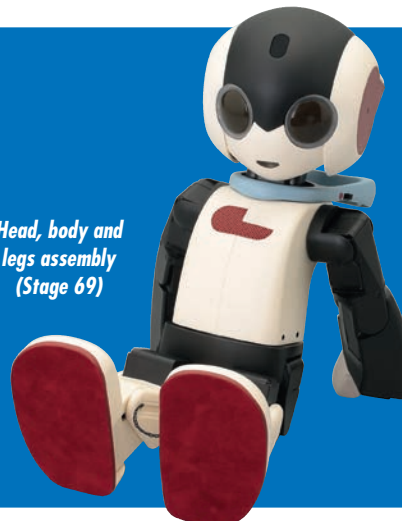


YOUR PARTS

- 1 Back cover
- 2 Left hand
- 3 Right hand
- 4 White adhesive strips
- 5 Black adhesive strips
- 6 Final micro SD card
- 7 4 x 1mm plastic washers x 6
- 8 M2 x 5mm countersunk screws x 5 (one is a spare)

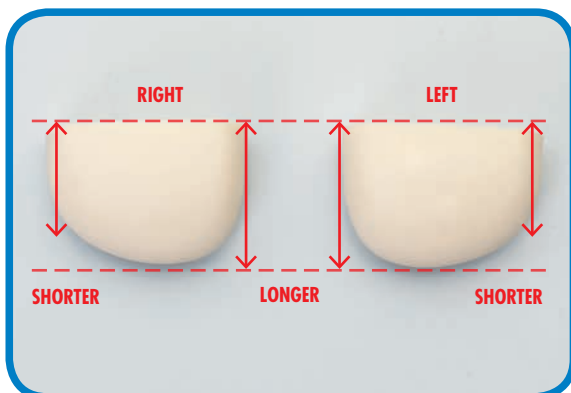
SAVED PARTS

Head, body and
legs assembly
(Stage 69)

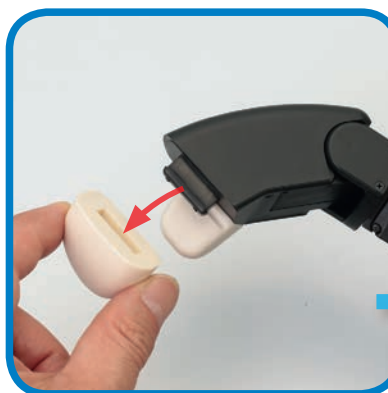


Chest stickers
(Stage 69)

COMPLETE ROBI'S HANDS



1 Familiarise yourself with the shape of Robi's left and right hands, using the diagram above.



2 Push the right hand onto the lip at the end of the right forearm. It should click into place.



3 Make sure the parts are sitting flush with one another.



4 Repeat for the left hand and forearm.



COMPLETE ROBI'S LEGS



5 The servo cables running from Robi's feet servos to his knees need to be hidden. You will begin with the right leg.



6 Gently work the cable into both holes, so that it is flush to the groove.



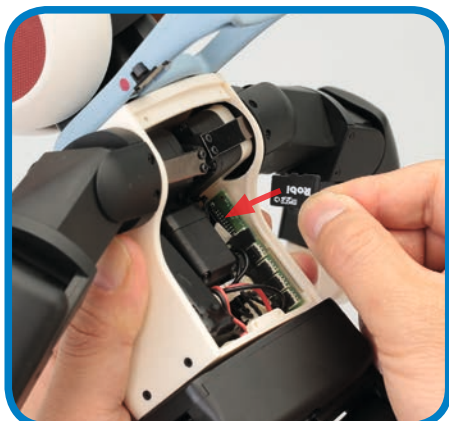


7 Cover the cable with the one of the white adhesive strips supplied with this stage.

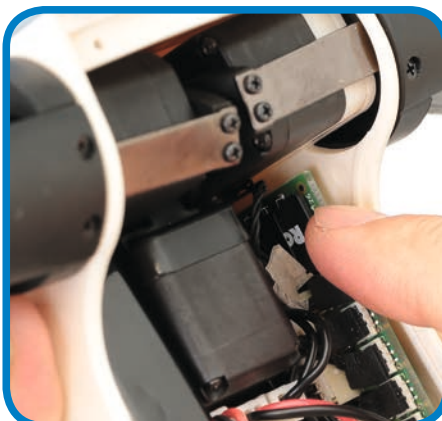


8 Smooth the strip out, then repeat for the other leg.

INSERT THE FINAL MICRO SD CARD



9 Carefully insert the final micro SD card supplied with this stage into the card slot on the CPU.



10 Make sure the card is correctly oriented as shown, then push it into place fully.

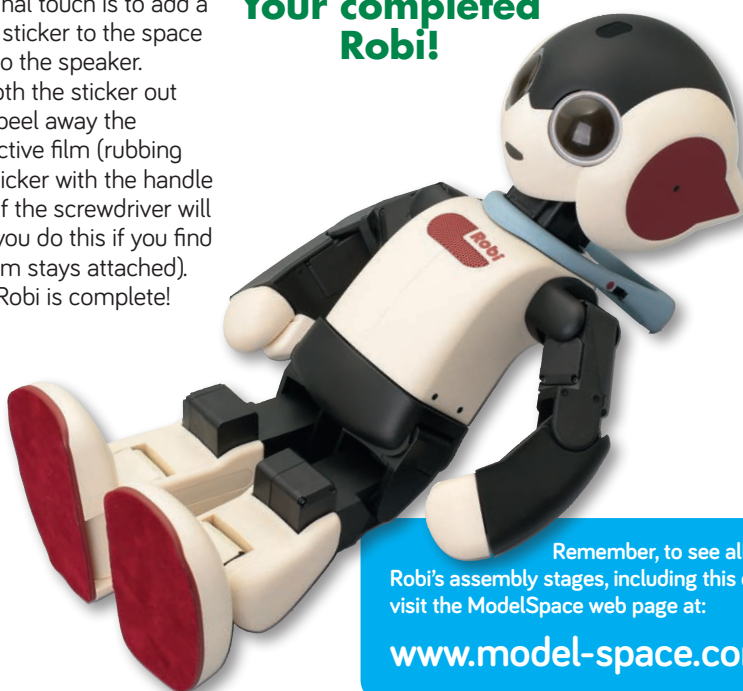


11 Cover Robi's back, then secure the cover with four M2 x 5mm countersunk screws.



12 The final touch is to add a 'Robi' sticker to the space next to the speaker. Smooth the sticker out then peel away the protective film (rubbing the sticker with the handle end of the screwdriver will help you do this if you find the film stays attached). Your Robi is complete!

Your completed Robi!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

START-UP AND MAINTENANCE

In this section, we show you how to bring your Robi to life, and look after him!

STARTING ROBI FOR THE FIRST TIME

Before turning Robi on for the first time, make sure the battery is fully charged and that the mains power is disconnected. Then, lay Robi down as shown below.

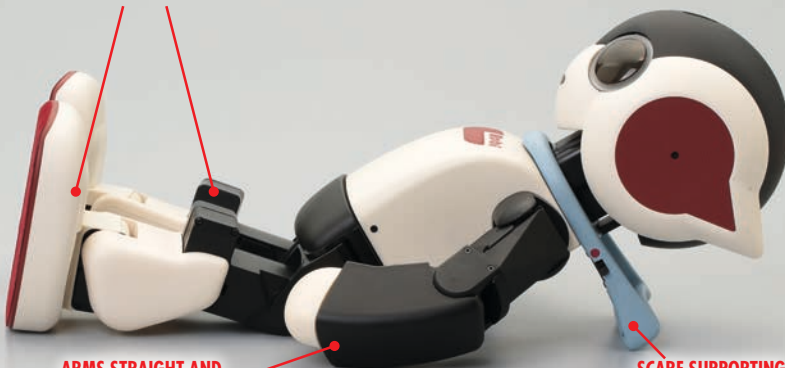
Steps 1-5 in this section only refer to the first time you turn Robi on.



LEGS STRAIGHT WITH
FEET AT RIGHT ANGLES

ARMS STRAIGHT AND
BY SIDES

SCARF SUPPORTING
FROM NECK



ROBI WAKES UP!

CLOSE-UP

When you turn your Robi on for the first time, he will go through a unique 'waking up' process. He will do this only once, so if you have been building your Robi with friends or family, you may want to make sure everyone is with you when you turn him on for the first time.

1

Lay Robi out on a flat, smooth surface, such as the floor or a tabletop. Make sure his limbs are positioned as shown.



2

Turn on Robi via his scarf switch.

3

After a few moments, Robi will begin to come alive!

ANSWER EACH QUESTION
WITH A CLEARLY-SPOKEN
'YES' OR 'NO'



4

Robi will sit up and ask you a series of Yes or No questions, to help determine his character. **Answer each with a clearly spoken 'yes' or 'no', speaking from about 50cm away from him.** Wait for Robi to finish speaking before giving your reply, and if he does not respond, repeat your answer clearly.

ROBI'S EYES WILL PULSATE,
AND HIS FEET MOVE
SLIGHTLY AS HE WAITS
FOR AN ANSWER



5

Answer all of Robi's questions until he announces that the process is complete, then you are ready to start interacting with Robi normally. See the *Living with Robi* guide supplied with this pack for information on the many ways you can talk to and play with your Robi!



Robi's personality will be defined by the answers you give, making sure your Robi is a good match for you. This is done only once, but don't worry: all of Robi's features will work, whichever personality of his is decided.

STARTING ROBI FOR THE SECOND TIME

Once Robi's personality is set, and after you switch him on for the first time, you no longer need to lay him out on his back. Robi can be switched on while standing, when sitting on a flat surface or his charging chair. Always make sure he is stable and balanced before switching his power on.



Remember, always hold Robi by his scarf when moving him!

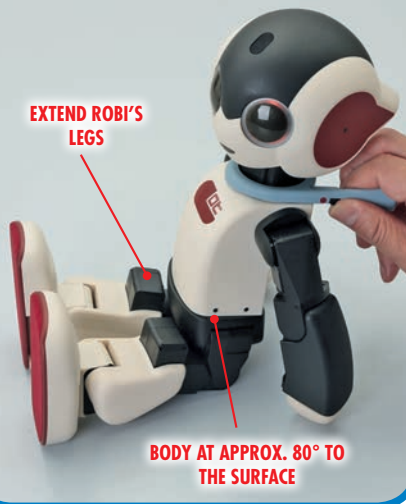
STARTING FROM A STANDING POSITION



6

Make sure Robi is standing on a smooth, flat surface in a stable and balanced position. He should be able to stand up without you holding him by the scarf. Then, holding him level by the scarf again, turn the power switch ON. Robi's servos should move to their neutral position; when he is set correctly you can let go of the scarf.

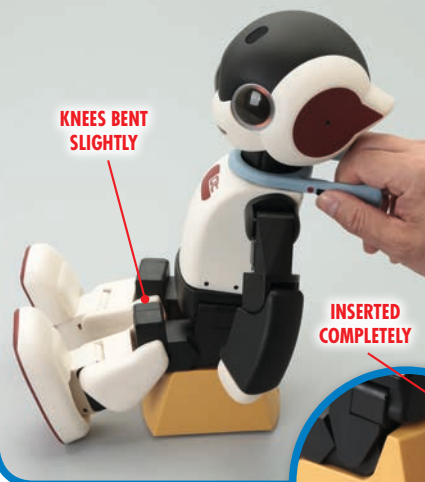
STARTING FROM A SITTING POSITION



7

When starting Robi from a sitting position, make sure his limbs are positioned as shown above, with his legs extending straight out before him and his feet perpendicular to these. His body should be almost perpendicular to the surface (approx. 80°) and his arms should support his body, which leans slightly backwards, resting on his hands. Again, hold Robi's scarf until his power is ON.

STARTING FROM THE CHARGING CHAIR



8

Robi can also be switched on while sitting on his charging chair, **but make sure the chair itself is not still connected to mains power – Robi must never be switched on while charging.** This time, Robi's legs should be slightly bent at the knees and again, hold Robi's scarf until his power is ON.

ROBI SAFETY AND USAGE TIPS

The *Living with Robi* guide supplied with this pack will give you all the information you need to be able to interact and play with Robi, but this page lists the key points on safety and usage.

Robi start-up and general use

- Robi moves whether he is standing up or sitting down, so always make sure there is enough space around him for him to do so safely. For example, be careful not to place Robi too close to the edge of a tabletop.
- Make sure there are no obstacles around Robi that he may bump into.
- Robi needs a smooth, flat surface to operate on. Surfaces like carpets or tablecloths can cause friction and disturb his movement and balance.
- Robi can stand up from a sitting position on his own, but if on an uneven surface, you may want to hold his scarf for extra support.
- You should never turn the power ON while Robi is charging. If he is on his charging chair and he is switched on, he will tell you to turn him off, remove him from the chair, and turn him on again.
- Do not use Robi while he is charging.

Robi voice recognition

- Robi's voice-recognition technology works best in a quiet environment with little background noise. In ideal conditions he will be able to hear and understand commands from a few metres.
- Robi's microphone for picking up commands is set into the front of his head, so if he has difficulty registering speech, move closer and direct the commands at his forehead.
- Talk to Robi in a clear tone at normal speed – speaking too fast or too softly will affect his ability to register words.
- It may take a few seconds for Robi to reply, depending on the type of command given. Give Robi time to answer, and if some seconds pass without reply, repeat the command.
- A moment of silence is needed before and after each command for the technology to register it successfully.
- Prompts and commands should be delivered exactly as they appear in the *Living with Robi* guide: speak in a clear, level tone with no pauses, and do not alter the wording at all.
- Any deviation from the above will reduce Robi's ability to recognise prompts.

The battery and automatic power-saving mode

- Operation time with a fully charged battery is up to 20 minutes, depending on the types of activities performed.
- If Robi is left standing for some time, he will sit down to conserve power.
- After three minutes of sitting, if no commands are received, Robi will power down to save battery. Turn the power switch to OFF at this point.
- Robi will warn you when his battery is running low, but you can also ask him how much power he has left – see the *Living with Robi* guide for more on this.

- Robi has many 'special functions' – such as quizzes and games – where he will wait for a specific response from you (all of which are explained in full in the *Living with Robi* guide). If he is left too long in this mode, his battery may run out, causing him to topple over. You can turn him off at his switch to avoid this.
- Only use the mains adapter or charging seat provided with the *Build Your Own Robi* series to charge your Robi.
- Do not remove the battery or tamper with it in any way.
- Keep the battery away from heat sources.
- Never handle the battery with wet hands, and keep it away from any damp areas. (See Stage 38 for more on battery care and use.)

Robi behaviour triggers

- In response to some greetings, Robi will extend his arm for a 'handshake'. When he does this, press down gently on his forearm.
- Robi has a timer mode, but this should not be used for important tasks.
- During some conversations Robi will talk of 'swimming', 'bathing' or firing a laser beam. These are all purely conversational: he must never be put in contact with water, and of course is equipped with no laser technology!
- Similarly, when Robi talks of 'calling security', no actual call will be made.

Other actions/functionality

- Only ever move Robi using his scarf – this is especially important when the power is on.
- It is natural for Robi's CPU, battery and servo motors to warm up a little while in use, but if these heat up too dramatically or rapidly, turn Robi OFF immediately.
- It is possible that a slight hissing sound will be heard as Robi speaks – this is completely normal.
- If Robi falls while walking or balancing, pick him up by the scarf and reset his stance. Also ensure the surface you are using is completely level and smooth.
- More on Robi's balancing is found later in this section.
- Keep Robi out of direct sunlight.
- Do not leave Robi in the standing position when switched off, and if leaving him unused for some time, make sure to store him where there is no stress on his joints, for example in the sitting or lying down position.
- Robi should not be in contact with other objects for long periods of time.
- If Robi's plastic exoskeleton becomes dirty, wipe it with a very slightly damp cloth, then rub dry immediately. A mild diluted detergent can be used to dampen the cloth, but on no account should Robi be immersed in water or made too wet during cleaning.

ROBI'S POSTURE AND SERVOS

Viewed from the side

Check that all of your Robi's limbs match those of this diagram.



TOP OF BODY PANEL AND
SHOULDER ARE PARALLEL

SIDE SCREW VISIBLE AHEAD OF FOREARM

HAND SLIGHTLY AHEAD OF
UPPER LEG/HIP

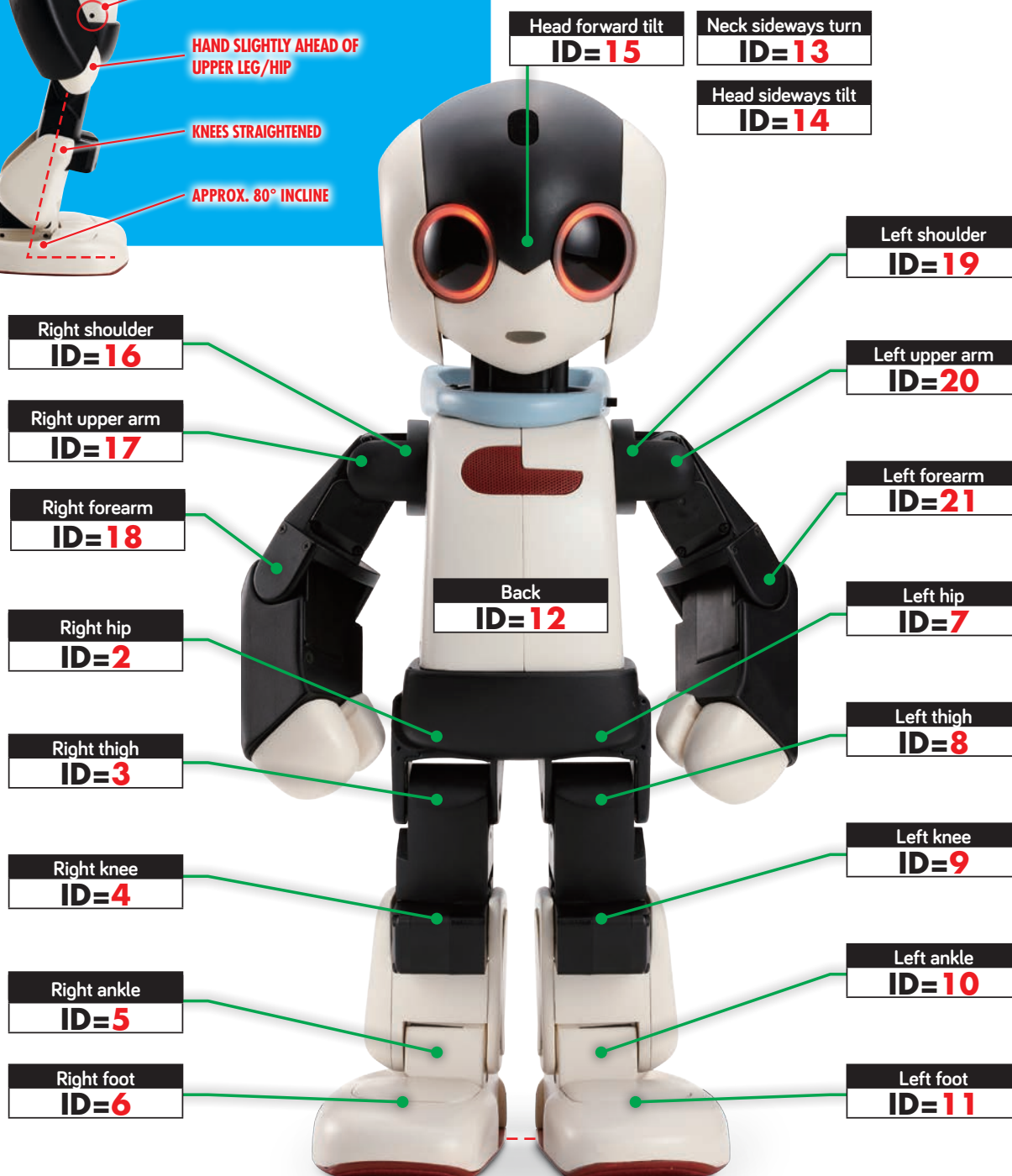
KNEES STRAIGHTENED

APPROX. 80° INCLINE

On this page we show the correct posture for Robi when standing, and list again each of the servo numbers that you have set over the previous packs. Use this page as a reference should you need to adjust or reset any of Robi's components, instructions for which are given over the last pages of this pack.

Viewed from the front

Check from the front that Robi's arms and legs match those of the diagram below. This is especially important for the legs, as any deviation from the intended neutral standing position can result in a loss of balance when moving.



DISTANCE BETWEEN HEELS SHOULD BE 2CM
DISTANCE BETWEEN TOES SHOULD BE 3CM

CORRECTING BALANCE ISSUES

If Robi falls over when walking or performing other physical activities, first check that the surface he is operating on is perfectly level and smooth. If this is not the issue, you may have to adjust his legs or feet, instructions for which follow.

- Place a ball, such as the one supplied in this series, on the tabletop. If it rolls, the tabletop is not level.
- Make sure that the surface is smooth.
- If Robi always falls on the same side, there is very likely a tilt on that side of the walking surface.

Check that Robi is assembled correctly

1. The power cord



Check that the red and black power cord is seated in the groove in the underside of Robi's hip panel. If it slips out from this groove, it can obstruct the movement of the legs, affecting Robi's balance.

2. The soles of Robi's feet



The soles of Robi's feet must be smooth and perfectly flat. If any small object is stuck to one of the soles, remove it; if the object is beneath the velvet pad, you will need to carefully peel this back to remove it. If one of the screw heads is protruding from the sole of the foot, tighten it so that it fits as it should.

3. The screws and cable routing



Check that the screws in Robi's legs circled above are fully tightened. If these screws become loose, the smooth operation of the leg joints (arrowed) can be affected. If this happens, there is also a chance that the servo cables routed through each leg can become caught in the joints and trapped, which also inhibits movement, so check for this too.

(The D-cuts)



You have been checking Robi's D-cut joints throughout the build, with the troubleshooting guides showing you how to reset these where necessary. However, check again now that the D-cut joints are set as they should be. The troubleshooting guides dealing with Robi's D-cut joints are found in Stages 52, 66 and 69.



The following steps show how to address any balance problems Robi may be experiencing that are *not* caused by misaligned D-cuts. Each of Robi's D-cut joints has been tested over the previous packs, so before proceeding with the below, double-check that Robi adequately performs the tests of Stages 52, 66 and 69. If he does, the problem may lie elsewhere, so follow the steps on this and the next page.

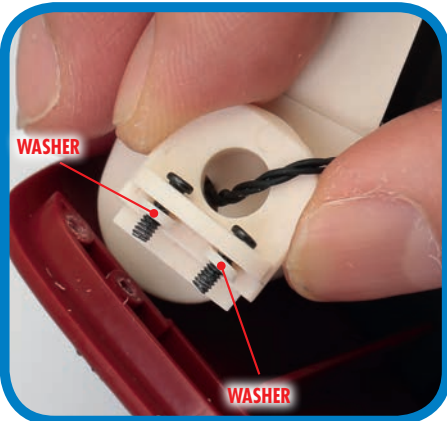
Adjusting the height of Robi's soles

1. If Robi falls backwards



1 If Robi is tilting backwards, then falling, when standing or walking, you will need to raise the height of the rear of his feet slightly. Begin by carefully peeling away the velvet pad and taking out all three circled screws to remove the foot side cover panel. The examples on this page show the right foot, but the process is the same for both feet.

2 Now loosen the two screws securing the ankle joint to the sole.



3 Now take two of the plastic washers supplied with this stage and fit them onto the threaded ends of the screws, as shown above. Tighten both screws back into the sole, so that the ankle joint is secure, but sitting slightly higher than it was previously.

4 Next, refit the rearmost screw from the underside of the sole, and place another washer on its threaded tip, on the inside of the sole.

2. If Robi falls forward

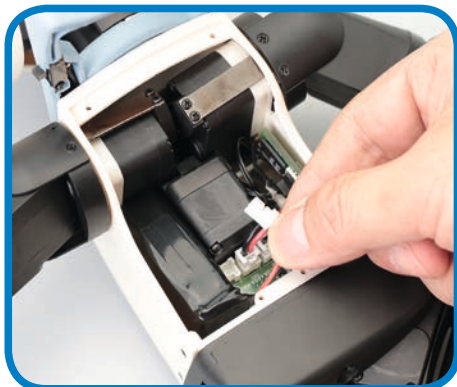


5 If Robi falls forward, the front of his feet require attention. Begin by repeating Steps 1 and 2, before also removing the screws holding the foot servo and cover to the sole. The right foot is used in this example.

6 Take two of the protective pads you have been using for the servo connectors and stick these to the indicated face of the plastic frame encasing the foot servo. Then rebuild the foot as before.

TAPING THE SHOULDER JOINTS

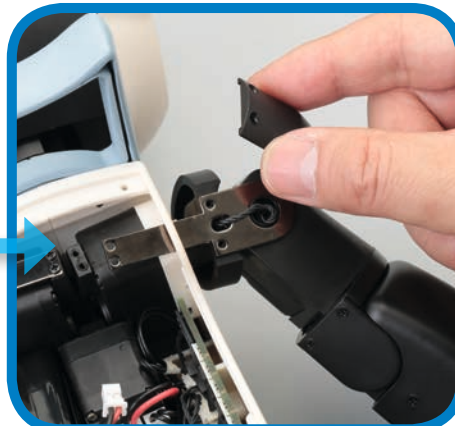
Over time, you may find Robi's limbs tremble a little as they move, which may affect Robi's overall movement and balance. To reduce this trembling, the adhesive tape supplied with this stage can be applied on the inside of Robi's body.



- 1** Remove Robi's back cover and disconnect the battery to prevent accidentally turning the power on.



- 2** Take out the four circled screws and remove the shoulder panel. The examples shown use the right arm, but the procedure is the same for both.



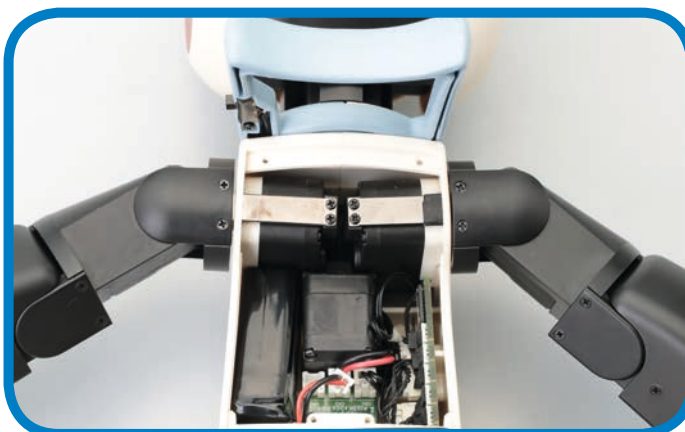
- 3** Gently slacken the servo cables so you can remove the metal joint from Robi's body.



- 4** Remove the backing from one of the strips of adhesive tape supplied with this stage and wrap it around the base of the joint.



- 5** Rebuild Robi's arm as before.



That completes the assembly of your Robi!
For more information on playing with and looking after Robi, see the *Living with Robi* guide, supplied with this pack.

LIVING WITH ROBI GUIDE

Robi is now complete!

Your *Living with Robi* guide will give you all the information you need to make the most of your very own Robi! In the guide, you will find directions on how to play games with Robi, how to have a conversation with him, and details of all the different ways to enjoy your new robot companion.

